

$$3.2. \langle f, g \rangle = \int_1^5 f(x) g(x) dx$$

$$a) u_1 = \frac{1}{\|1\|} = \frac{1}{2} \quad \int_1^5 \frac{x}{2} dx = \left[\frac{x^2}{4} \right]_1^5 = \frac{24}{4} = 6$$

$$u_2 = \frac{x - \langle \frac{1}{2}, x \rangle \frac{1}{2}}{\| \dots \|} = \frac{x - 3}{\|x - 3\|} = \frac{\sqrt{3}}{4}(x - 3) \quad \int_1^5 (x - 3)^2 dx = \left[\frac{(x - 3)^3}{3} \right]_1^5 =$$

$$u_3 = \frac{x^2 - \langle \frac{\sqrt{3}}{4}(x - 3), x^2 \rangle \frac{\sqrt{3}}{4}(x - 3) - \langle \frac{1}{2}, x^2 \rangle \frac{1}{2}}{\| \dots \|} = \frac{8}{3} + \frac{8}{3} = \frac{16}{3}$$

$$= \frac{x^2 - \frac{3}{16} \cdot 32(x - 3) - \frac{1}{4} \cdot \frac{124}{3}}{\| \dots \|} =$$

$$\int_1^5 (x - 3)x^2 dx = \left[\frac{x^4}{4} - x^3 \right]_1^5 = \frac{625}{4} - 125 - \frac{1}{4} + 1 = 156 - 124 = 32$$

$$\int_1^5 x^2 dx = \left[\frac{x^3}{3} \right]_1^5 = \frac{124}{3}$$

$$= \frac{x^2 - 6x + 18 - \frac{31}{3}}{\| \dots \|} =$$

$$= \frac{x^2 - 6x + \frac{23}{3}}{\| \dots \|} = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right)$$

$$\mathcal{B}_{ON} = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x - 3), \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \right)$$

$$b) \mathcal{B} \text{ je ON} \Rightarrow G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_n$$

$$[\sin x]_{\mathcal{B}} = \begin{pmatrix} \langle \frac{1}{2}, \sin x \rangle \\ \langle \frac{\sqrt{3}}{4}(x - 3), \sin x \rangle \\ \langle \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right), \sin x \rangle \end{pmatrix} = \begin{pmatrix} 0,128 \\ \frac{\sqrt{3}}{4}(-3,448) \\ \frac{3\sqrt{5}}{16}(-0,299) \end{pmatrix}$$

$$\sin x = 0,064 - 3,448 \cdot \frac{3}{16}(x - 3) - \frac{45}{256} \cdot 0,299 \left(x^2 - 6x + \frac{23}{3} \right)$$

$$\sin x = 0,064 - 0,647x + 3,01647 - 0,053x^2 + 6 \cdot 0,053x - \frac{23}{3} \cdot 0,053$$

$$\sin x = 1,599 - 0,329x - 0,053x^2$$

$$c) \quad ||\sin x - ax^2 - bx - c|| = \sqrt{\int_1^5 (\sin x - ax^2 - bx - c)^2 dx} = *$$

$$\int_1^5 \sin^2 x dx = 2.363$$

$$\int_1^5 -2(ax^2 + bx + c)\sin x dx = -2 \left(a \int_1^5 x^2 \sin x dx + b \int_1^5 x \sin x dx + c \int_1^5 \sin x dx \right) =$$

$$= -2(-18.337a - 2.678b + 0.257c)$$

$$\int_1^5 (ax^2 + bx + c)^2 dx = a^2 \int_1^5 x^4 dx + 2a \int_1^5 x^2(bx + c) dx + \int_1^5 (bx + c)^2 dx =$$

$$= a^2 \int_1^5 x^4 dx + 2a \left(b \int_1^5 x^3 dx + c \int_1^5 x^2 dx \right) + b^2 \int_1^5 x^2 dx + 2bc \int_1^5 x dx + c^2 \int_1^5 1 dx =$$

$$= 624.8 a^2 + 2ab \cdot 156 + 2ac \cdot 41.333 + 41.333 b^2 + 24bc + 4c^2 =$$

$$* \quad \sqrt{2.363 + 36.674a + 5.356b - 0.514c + 624.8a^2 + 312ab + 82.666ac}$$

$$\sqrt{41.333b^2 + 24bc + 4c^2} = \underline{\underline{0.317}}$$